

Clinical evaluation of combined programmed cell death protein 1 inhibitor and poly(ADP-ribose) polymerase inhibitor in metastatic castration-resistant prostate cancer patients: Insights from a real-world study

Yuan Shao^{a,b}, Zihao Liu^{a,b,c}, Yinchu Zhang^{a,b}, Yang Liu^{a,b}, Hua Huang^{a,b}, Zhinan Fu^{a,b}, Zhen Yang^{a,b}, Zeyuan Wang^{a,b}, Shen Zhang^{a,b}, Yong Wang^{a,b,*}

^aDepartment of Urology, The Second Hospital of Tianjin Medical University, Tianjin, China; ^bTianjin Institute of Urology, The Second Hospital of Tianjin Medical University, Tianjin, China; ^cSchool of Pharmaceutical Science and Technology, Tianjin University, Tianjin, China.

Abstract

Background: This study aimed to evaluate the clinical efficacy and safety of combining a programmed cell death protein 1 (PD-1) inhibitor with a poly(ADP-ribose) polymerase inhibitor (PARPi) in patients with metastatic castration-resistant prostate cancer (mCRPC) who progressed after multiple lines of treatment, from a real-world perspective.

Materials and methods: This open-label, single-arm, prospective study enrolled patients with mCRPC who had experienced disease progression after docetaxel and at least 2 lines of next-generation hormonal agents to receive camrelizumab (PD-1 inhibitor) and fluzoparib (PARPi). The primary endpoints were radiographic progression-free survival and overall survival (OS), and the secondary endpoints were prostate-specific antigen progression-free survival and safety.

Results: Eight patients with mCRPC who met the inclusion criteria were enrolled. The results showed that the median radiographic progression-free survival was 5.1 months, the median OS was 8.1 months, and the median prostate-specific antigen progression-free survival was 3.1 months. Safety analysis revealed that 87.5% of the patients experienced one or more treatment-related adverse events (AEs), with 37.5% reporting grade 3 or higher treatment-related AEs. None of the patients discontinued treatment because of treatment-related AEs.

Conclusions: This real-world study demonstrated that the combination of a PD-1 inhibitor and PARPi exhibited sustained antitumor activity with an acceptable safety profile in the fourth-line treatment of patients with mCRPC.

Keywords: Metastatic castration-resistant prostate cancer; Poly(ADP-ribose) polymerase inhibitor; Programmed cell death protein 1 inhibitor; Real-world study; Biomarker

1. Introduction

Metastatic castration-resistant prostate cancer (mCRPC) is a lethal stage of prostate cancer that is characterized by continued disease progression despite androgen deprivation therapy.^[1] Although next-generation hormonal agents (NHAs) and docetaxel improve survival, most patients eventually experience disease progression after multiple lines of treatment, with limited therapeutic options remaining.^[2] Therefore, there is an urgent need to develop novel and effective treatment strategies for improving the survival of patients with mCRPC.

Poly(ADP-ribose) polymerase inhibitors (PARPis) have demonstrated survival benefits in patients with mCRPC harboring homologous recombination repair mutations (HRRMs), particularly BRCA1/2 alterations.^[3] Meanwhile, immune checkpoint inhibitors (ICIs), such as programmed cell death protein 1 (PD-1) inhibitors, have shown limited efficacy in mCRPC owing to the immunosuppressive tumor microenvironment. Preclinical studies have suggested that PARPi can enhance immunogenicity and sensitize tumors to ICIs via multiple mechanisms.^[4]

This study aimed to evaluate the clinical efficacy and safety of a PD-1 inhibitor plus PARPi in patients with mCRPC who had failed multiple lines of treatment, including NHA and docetaxel. In addition, we explored potential biomarkers to guide future clinical decision making.

2. Materials and methods

This open-label, single-arm, prospective study was conducted at The Second Hospital of Tianjin Medical University and registered in the Chinese Clinical Trial Registry (ChiCTR2100043664). Eligible patients were 18 years or older and had histologically confirmed mCRPC, Eastern Cooperative Oncology Group performance status ≤ 2 , and prior treatment with docetaxel and at least 2 NHAs. The

*Corresponding Author: Yong Wang, Department of Urology, The Second Hospital of Tianjin Medical University, Tianjin, 300211, China; Tianjin Institute of Urology, The Second Hospital of Tianjin Medical University, Tianjin, 300211, China. Email address: wy@tmu.edu.cn.

Yuan Shao, Zihao Liu, and Yinchu Zhang contributed equally to this work.

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patients had not previously received PD-1 inhibitors or PARPi and underwent targeted next-generation sequencing of circulating tumor DNA from peripheral blood using an 808-gene panel.

Patients received camrelizumab (200 mg intravenously every 3 weeks) combined with fluzoparib (150 mg orally twice daily), irrespective of their HRRm status. Treatment was continued until disease progression, intolerable toxicity, or withdrawal of consent occurred. All the patients underwent castration via androgen deprivation therapy or prior orchiectomy.

Imaging was performed every 12 weeks (including computed tomography of the chest, abdomen, and pelvis; magnetic resonance imaging of the prostate; and radionuclide bone scintigraphy), and prostate-specific antigen (PSA) levels were monitored every 4 to 6 weeks. Adverse events (AEs) were recorded according to the National Cancer Institute Common Terminology Criteria for Adverse Events, version 5.0 (CTCAE 5.0).

The primary endpoints were radiographic progression-free survival (rPFS) and overall survival (OS). The secondary endpoints included PSA progression-free survival (PSA-PFS) and safety. Kaplan-Meier and log-rank tests were used for survival analysis. Statistical significance was set at $p < 0.05$. All analyses were performed using R software (version 4.2.0; R Foundation for Statistical Computing, Vienna, Austria).

3. Results

3.1. Baseline characteristics and efficacy

Eight mCRPC patients were enrolled in this study. The median age was 70.5 years. The median time from mCRPC diagnosis to the initiation of PD-1 inhibitor plus PARPi therapy was 17.3 months, with a median of 5 treatment cycles administered. The median follow-up was 6.0 months (Supplemental Digital Content, Supplementary Table S1). The median rPFS was 5.1 months (Fig. 1A). Subgroup analysis showed that PSA, hemoglobin, and lymphocyte levels prior to combination therapy were associated with rPFS. The median OS was 8.1 months (Fig. 1B). PSA and alkaline phosphatase levels before treatment were significantly associated with OS. The median PSA-PFS was 3.1 months (Fig. 1C), and the PSA level before treatment was also associated with PSA-PFS.

3.2. Exploratory analysis

Targeted gene sequencing revealed that 3 patients (38%) had mutations in CDK12 or TP53, and 2 patients (25%) had aberrations in AR. In addition, 5 patients (62.5%) had one or more mutations in HRR genes. None of the patients exhibited high microsatellite

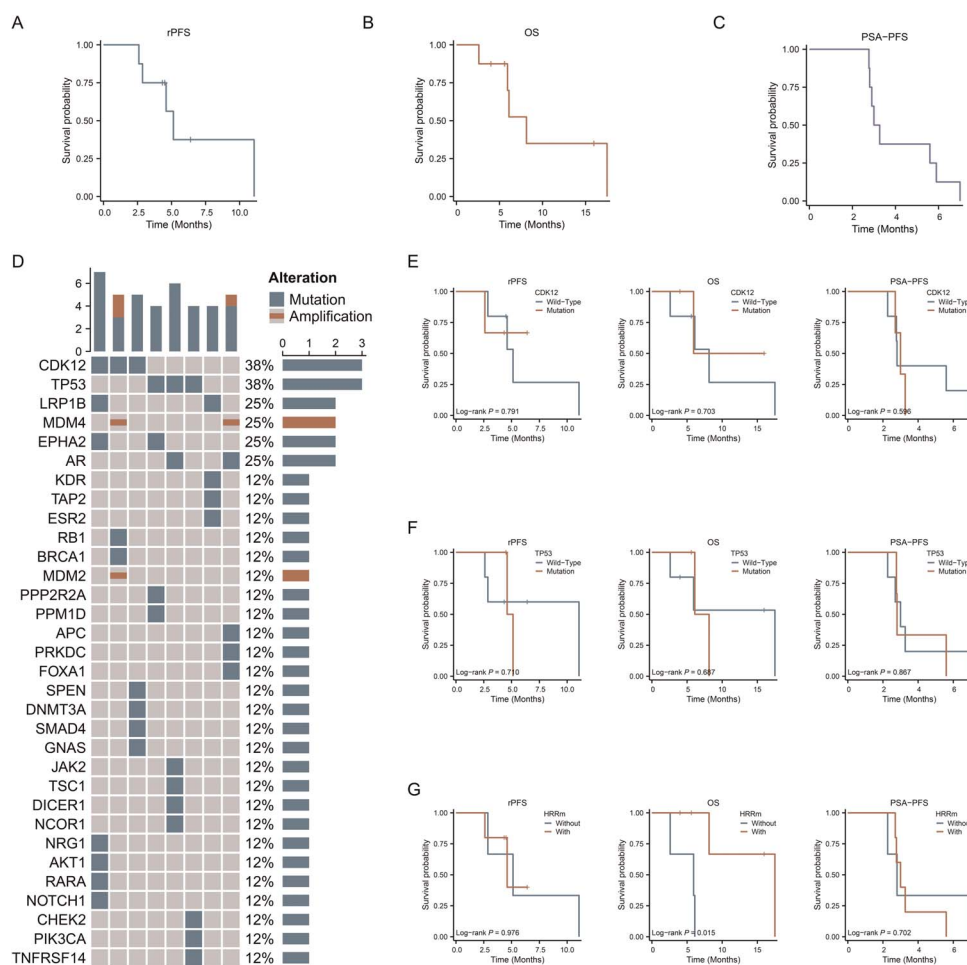


Figure 1. Efficacy and exploratory analysis of mCRPC patients treated with a PD-1 inhibitor and a PARPi. (A) Kaplan-Meier survival curves of rPFS; (B) Kaplan-Meier survival curves of OS; (C) Kaplan-Meier survival curves of PSA-PFS; (D) summary of genetic alterations identified in patients; (E) Kaplan-Meier survival curves of rPFS, OS, and PSA-PFS stratified by CDK12 mutation status; (F) Kaplan-Meier survival curves of rPFS, OS, and PSA-PFS stratified by TP53 mutation status; and (G) Kaplan-Meier survival curves of rPFS, OS, and PSA-PFS stratified by HRRm status. HRRm = homologous recombination repair mutation; mCRPC = metastatic castration-resistant prostate cancer; OS = overall survival; PARPi = poly(ADP-ribose) polymerase inhibitor; PD-1 = programmed cell death protein 1; PSA-PFS = prostate-specific antigen progression-free survival; rPFS = radiographic progression-free survival.

instability or mismatched repair deficiencies. Exploratory analysis indicated that patients with HRRms had longer OS (Fig. 1).

3.3. Safety

Treatment-related AEs occurred in 7 of 8 patients (87.5%), with 3 patients (37.5%) experiencing grade ≥ 3 AEs. The most common AEs included decreased hemoglobin level, leukopenia, neutropenia, loss of appetite, weight loss, fatigue, and diarrhea. Grade ≥ 3 anemia occurred in 2 patients and was managed with blood transfusion. No treatment-related deaths or treatment discontinuation were observed (Supplemental Digital Content, Supplementary Table S2).

4. Discussion

In the past few years, several clinical trials have been conducted to explore the applications of ICIs in patients with mCRPC. In the KEYNOTE-199 and CA184-043 trials, ICIs monotherapy showed limited antitumor activity.^[5,6] Because of the limited efficacy of ICIs monotherapy, multiple studies have evaluated ICI-based combination strategies in patients with mCRPC. In the KEYNOTE-921 and KEYNOTE-365 (cohort C) trials, ICIs combined with chemotherapy or androgen receptor pathway inhibitors exhibited antitumor activity, although the benefits remained modest.^[7,8] Therefore, these findings have led the current research to focus on achieving synergistic effects by combining ICIs with other treatment modalities for mCRPC.

Poly(ADP-ribose) polymerase inhibitor has shown efficacy in patients with HRRm, as demonstrated in the PROfound and PROpel trials. Theoretically, PARPi enhances sensitivity to ICIs by upregulating programmed cell death ligand 1, activating the cGAS-STING pathway, and promoting T cell-mediated cytotoxicity, especially in HRR-deficient tumors.^[4] As a result, the interaction between HRR pathway deficiencies and immune responses establishes the theoretical basis for combination therapy involving ICIs and PARPi. The KEYNOTE-365 (cohort A) and KEYLYNK-010 trials explored the therapeutic value of PD-1 inhibitor and PARPi combinations in second- and third-line mCRPC patients, respectively.^[9,10] However, patients with mCRPC who have failed multiple lines of systemic therapy often face limited treatment options and poor prognoses. In light of this clinical challenge, we explored the therapeutic effects of a PD-1 inhibitor plus a PARPi in the fourth-line treatment of patients with mCRPC in this real-world study. Our results showed that the median rPFS was 5.1 months, and the median OS was 8.1 months. Therefore, the combination of a PD-1 inhibitor and PARPi continues to demonstrate antitumor activity in patients with mCRPC who have failed multiple lines of treatment.

In addition to efficacy, 87.5% of patients experienced one or more treatment-related AEs, and 37.5% experienced at least 1 grade 3 or higher treatment-related AE in this real-world study. Hence, although combination therapy shows antitumor activity, its toxicity should not be overlooked. Prior to initiating combination therapy, a thorough evaluation of the potential clinical benefits and risks for each patient is necessary. During treatment, close monitoring is essential, and timely management of AEs is critical to avoid treatment interruption or discontinuation.

The limitations of this study include the small sample size, lack of a control group, and limited follow-up. These findings should be interpreted with caution and validated through larger prospective trials.

5. Conclusions

In conclusion, this real-world study demonstrated that combination therapy with a PD-1 inhibitor and PARPi demonstrates antitumor activity in patients with mCRPC who had received prior treatment with docetaxel and at least 2 NHAs. In addition, baseline PSA levels and HRRm were identified as potential prognostic biomarkers, providing valuable evidence for guiding future clinical decision making.

Acknowledgments

None.

Statement of ethics

The study was conducted in accordance with the Declaration of Helsinki and its subsequent revisions and was approved by the ethics committee of The Second Hospital of Tianjin Medical University (approval number: 2021-001). This clinical trial was registered with the Chinese Clinical Trial Registry under the registration number ChiCTR2100043664. Written informed consent was obtained from all participants prior to study initiation.

Conflict of interest statement

YW is a member of editorial board of *Current Urology* and confirms no involvement in any stage of this article's review process, ensuring unbiased editorial decision making. The other authors declare no conflicts of interest.

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Data availability

The original data presented in this study are included in the article and supplementary material. Further inquiries should be directed to the corresponding authors.

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